

Pranav Kumar Soma

SOFTWARE ENGINEER • MACHINE LEARNING • AI SYSTEMS

pranavsoma11@gmail.com • linkedin.com/in/pranav-kumar-soma • pranavsoma.me

San Francisco Bay Area, CA • Available January 2027

EDUCATION

M.S. Computer Science, University of California San Diego <i>GPA 4.00/4.00 • Jacobs School of Engineering • completed in 1 year</i>	Jan 2026 – Dec 2026 La Jolla, CA
B.S. Computer Science, University of California San Diego <i>GPA 3.96/4.00 • Cum Laude, CSE Department Honors, Provost Honors every quarter • completed in 2 years</i>	Sep 2023 – Dec 2025 La Jolla, CA

EXPERIENCE

Software Engineer Intern, ServiceNow <i>AI Voice Agent Analytics</i> - Shipped a full-stack conversation-history troubleshooting platform unifying 20+ data tables into per-call drill-down reports, live across 24+ enterprise instances serving 10K+ voice calls/day - Cut report load 7s to under 1s over millions of records via lazy loading and caching; authored PRDs, ran design reviews, presented to a VP and Senior Director. React, Playwright, Jenkins	Jun 2026 – Sep 2026 San Diego, CA
Software Engineer Intern, ServiceNow <i>AI Voice Agents • Technical Skills Award, top intern of 800+</i> - Owned the AWS Lex V1 to V2 migration end to end (8 intents, 48+ slots) ahead of end-of-life: 45% lower agent-response latency, validated live with an enterprise customer. Return offer - Built “Wrap Up,” an AI call-summarization flow (Seismic, data brokers, JavaScript) cutting average call-closing time 6 minutes across 10K calls/day	Jun 2025 – Sep 2025 San Diego, CA
AI Research Fellow & Agentic Systems Lead, Laboratory for Emerging Intelligence, UCSD - Lead 6 engineers on a Teradata-funded agentic workbench (100+ pilot users): 60+ MCP servers over all 600 UCSD APIs, a planner/executor split that degrades gracefully, and a skill-refinement loop tuning latency and token spend - Fine-tuned (LoRA, SFT, RLHF) and deployed the AI Tutor across UCSD, SDSU, and Cal Poly Pomona; a controlled study over 600+ students measured 64% fewer TA overtime hours, 87% higher engagement	Nov 2024 – Present
Graduate Researcher, Qualcomm Institute / Calit2, HXI Lab <i>AILA, the AI Living Assistant for the LifeSaver Project</i> Turned free-form patient interaction into symptom recognition and risk assessment over a MongoDB vector store of UMLS codes; fine-tuned local Qwen models on the SD Supercomputer Center. First-author poster	Nov 2024 – Dec 2025 San Diego, CA
Founding Software Engineer, Stealth Startup <i>Autonomous systems • part-time</i> Building the hardware-agnostic core operating system for an autonomy stack: multi-agent coordination across distributed edge nodes, real-time sensor fusion, low-latency C2 dashboards, hardware-in-the-loop simulation	Apr 2026 – Present
Honors & M.S. Thesis Researcher, UCSD CSE, Computational Redistricting <i>Advisors: Dr. Deepak Kumar, Dr. Imani Munyaka, Dr. Scott Desposato</i> Built 3 districting generators over Census 2020 data at 158K to 5.5M blocks/state (XGBoost + graph partitioning at MAE 0.042, Sequential Monte Carlo, greedy DP); reproduces official district counts for AL, CA, TX	Sep 2024 – Present La Jolla, CA

SELECTED PROJECTS

- ORCA, AI Music Production** (*Best Project, San Diego Tech Conference: 1 of 30 from 200+*) • OpenCV score reading, neural source separation, CNN audio-to-score; **98% arrangement accuracy**. Flask, Docker, React
- Spay.LA** • led **11 developers** as Engineering Manager on a MERN fundraising platform, **raised \$75K** • **TSE Constellation** • published React/TypeScript design system used org-wide by **100+** engineers

LEADERSHIP

- Co-Founder & President, Real World AI Network (2026–Present)** • **7,000+** student federation, **50+ faculty**, 6 schools.
- President, CSE Society @ UCSD (2025–2026)** • **300+** members, **21+** projects, sponsors **2 to 16**, **\$50K+** raised, first **Linux Foundation** hackathon.
- Co-Founder & EVP, University AI Alliance** • national network (MIT, Stanford, Caltech, CMU), **3,000+** students, a16z speedrun pipeline.

PUBLICATIONS & HONORS

- Publications:** ORCA (paper, SDTC 2025; poster, SAIRS 2025), THIA (SDTC 2025), VITON (SAIRS 2025), first-author AILA poster; redistricting in prep for ACM CHI.
- Honors:** ServiceNow Technical Skills Award • Cum Laude & CSE Department Honors • YC Startup School 2026 • National Merit Finalist • USACO Silver • AP Scholar with Distinction.

TECHNICAL SKILLS

- Languages:** Python, C++, C, Java, TypeScript, JavaScript, SQL, Bash, ARM Assembly, Swift, Kotlin
- ML & AI:** PyTorch, Transformers, scikit-learn, XGBoost, OpenCV; LLM fine-tuning (LoRA, SFT, RLHF), RAG and GraphRAG, vector search, LLM agents, MCP, A2A, LLM-as-judge evals, multimodal models, CNNs, LSTMs, RL, NLP/NLU, computer vision
- Systems & Data:** Docker, Kubernetes, Jenkins CI/CD, GCP, AWS (Lex, Lambda), Git, Linux, ROS; FastAPI, Flask, Node.js, Express, Django REST, React; MongoDB (vector), PostgreSQL, MySQL; NumPy, Pandas, GeoPandas, pymetis; multi-node GPU training, distributed and real-time systems, system design; Playwright, Postman, Agile